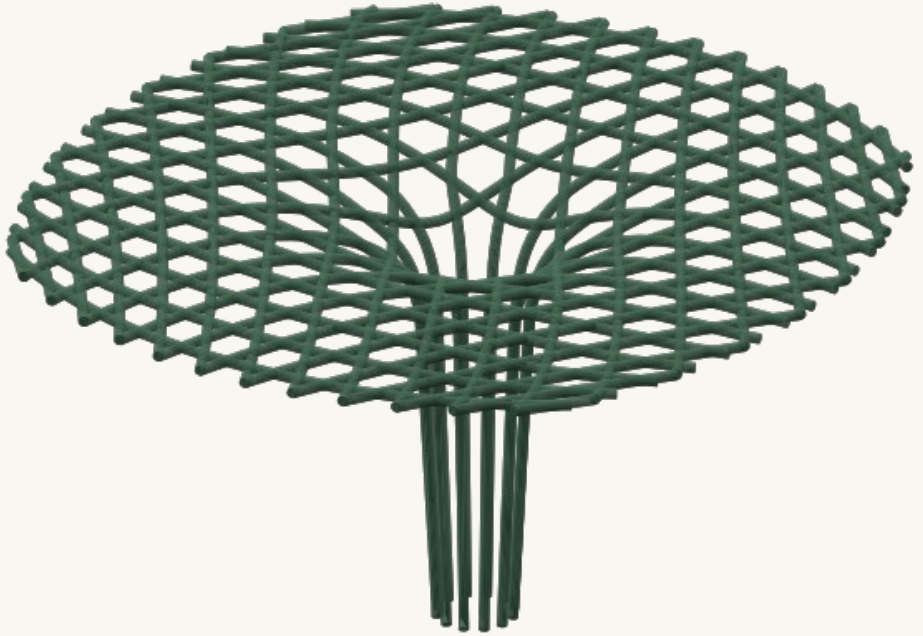
VineWeave Canopy

A tree-like climbing-plant support for pedestrian shade

Parametric design · Planting · Shade performance



TEAM & RESPONSIBILITIES

臧唯羽

Form & parametric modelling

林新淇

Materials & construction

虞寒哲

Shade performance & application

徐世程

PPT production & integration

陈淼

English presentation & Q&A

Translate the brief into five design decisions

A public shade facility must connect environmental intent with a buildable support.

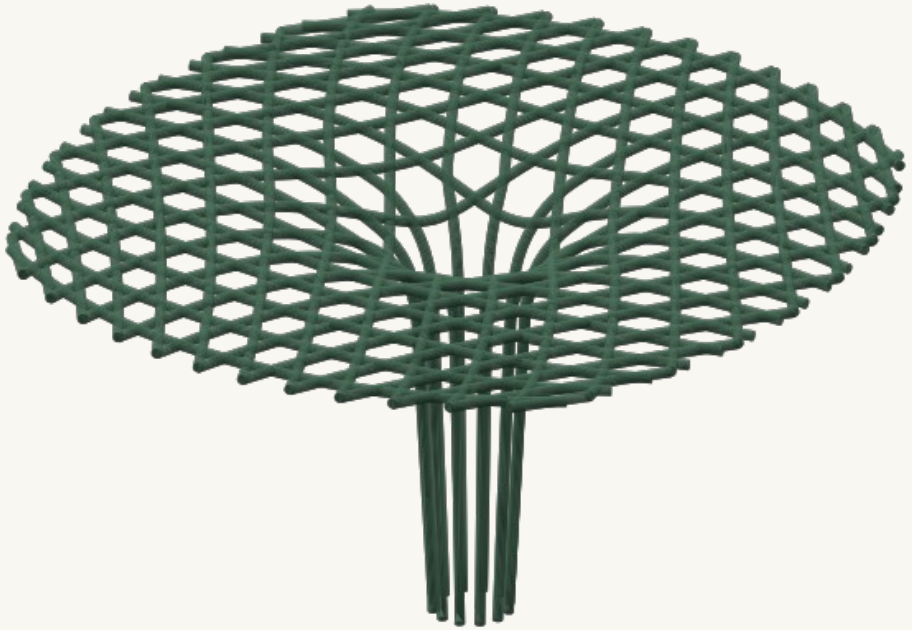
COURSE REQUIREMENT	OUR RESPONSE	WORK PRODUCED
Structure and materials	Steel supports + proposed bamboo crown	Material screen, member schedule and node concept
Geometric parameters	Control crown width, height and corners	Rhino/GH geometry, plans and parameter models
Shade and airflow	Broad crown + planting + open sides	Shadow projection; conceptual airflow analysis
Plant integration	Screen climbers and organise maintenance	Candidate screen, 2+1 planting proposal, cover scenarios
Assembly and use	Prefabricate, assemble and check occupied-hour shade	Build-up sequence, budget and rest-node application

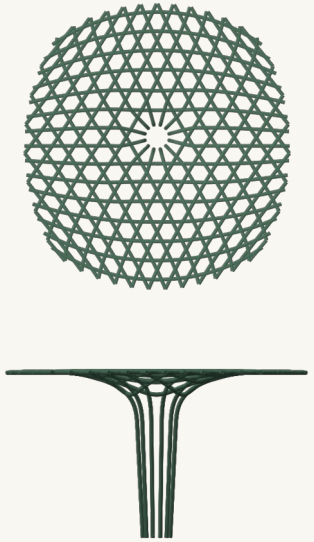
The design is assessed as a system—not as an isolated shape.

Shade/greening mechanisms: [1,2]. Project requirements and proposed strategy: P4.

From controlled centre lines to a measurable model

Rhino and Grasshopper provide a repeatable geometric basis for the analysis.





PLAN / ELEVATION

01 DEFINE CENTRE LINES

Height, span and crown controls

02 SHAPE THE CROWN

Corner factor 0.85; expansion 1.08

03 REBUILD MEMBERS

59 exterior pipe solids; radius 40 mm

04 EXTRACT & ANALYSE

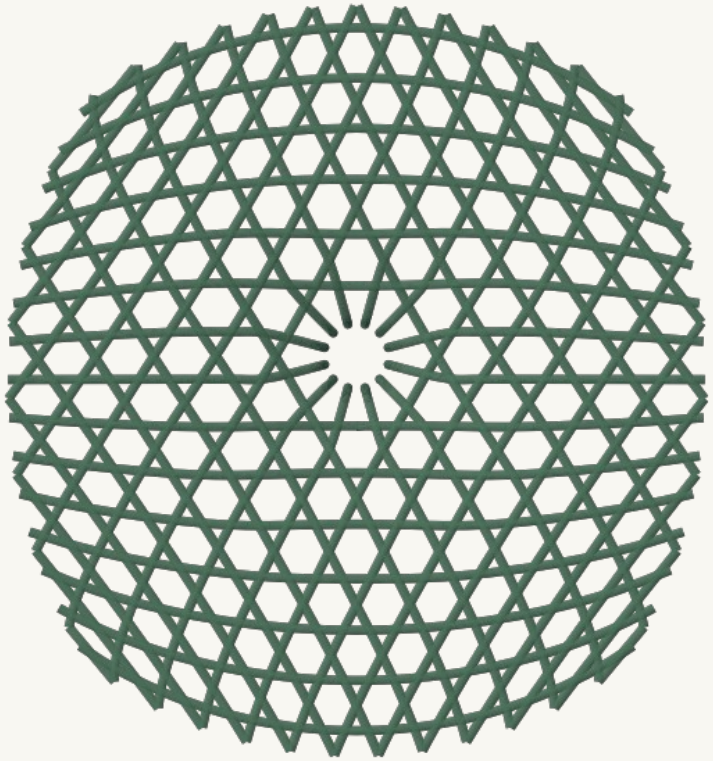
Geometry quantities and a leaf envelope

6.22 × 6.66 m PLAN	3.48 m MAXIMUM HEIGHT	59 MEMBERS

Project model, GH/C# method and parameter records (P1). Valid geometry is not a structural certification.

Let the crown, opening and edges perform different roles

The support is geometric; its future leaf cover is a separate design assumption.



01 BROAD INTERWOVEN CROWN

Distributes the proposed climbing envelope above the rest space.

02 CENTRAL LEAF OPENING

The analysis assumes an opening; vegetation could bridge it without pruning.

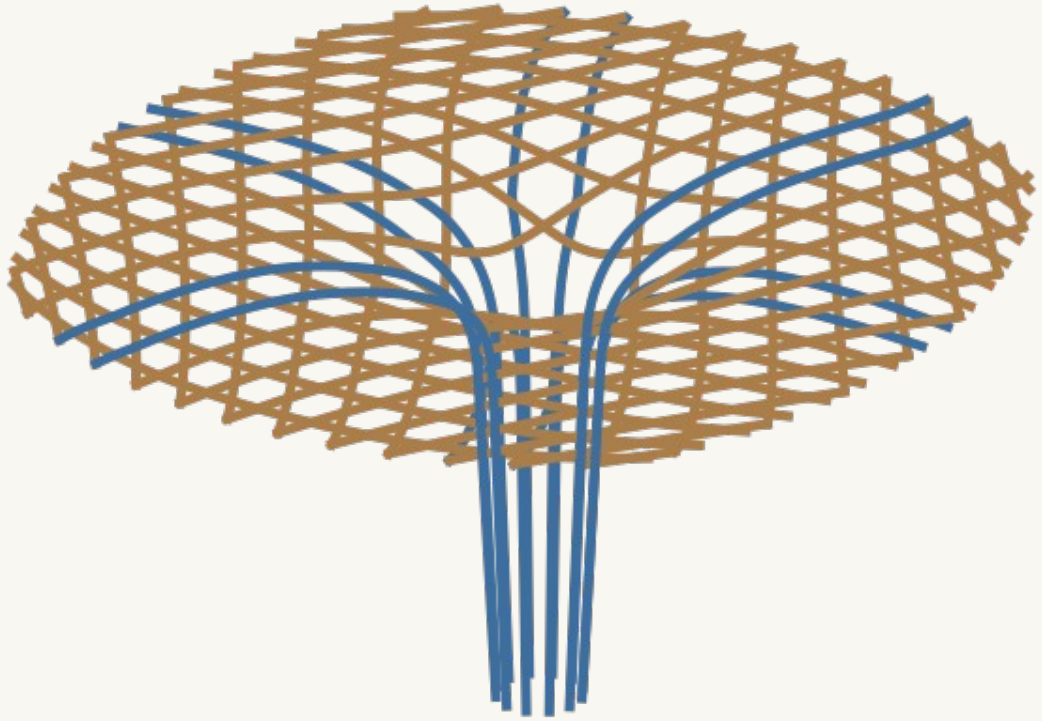
03 OPEN PEDESTRIAN EDGE

Keep movement, maintenance access and potential airflow paths around the support.

Project geometry and proposed planting envelope (P1, P4). Leaf openings depend on actual growth and pruning.

Separate the support role from the crown role

Classify members by whether their centre lines reach the ground.



ZONE	PROPOSED SECTION	LENGTH
12 steel supports	Q355B tube Ø80 × 4 mm	69.38 m
47 bamboo crown members	Engineered bamboo solid Ø80 mm	219.28 m
TOTAL	59 members	288.66 m

Steel supports address compression and wind bending. The bamboo crown is governed by stiffness, treatment and replaceability.

Project take-off and proposed member schedule (P1, P3). Sections and material mass remain preliminary specifications.

Screen strength, cost, carbon and outdoor care together

Representative workbook inputs explain the choice; no single material leads every criterion.

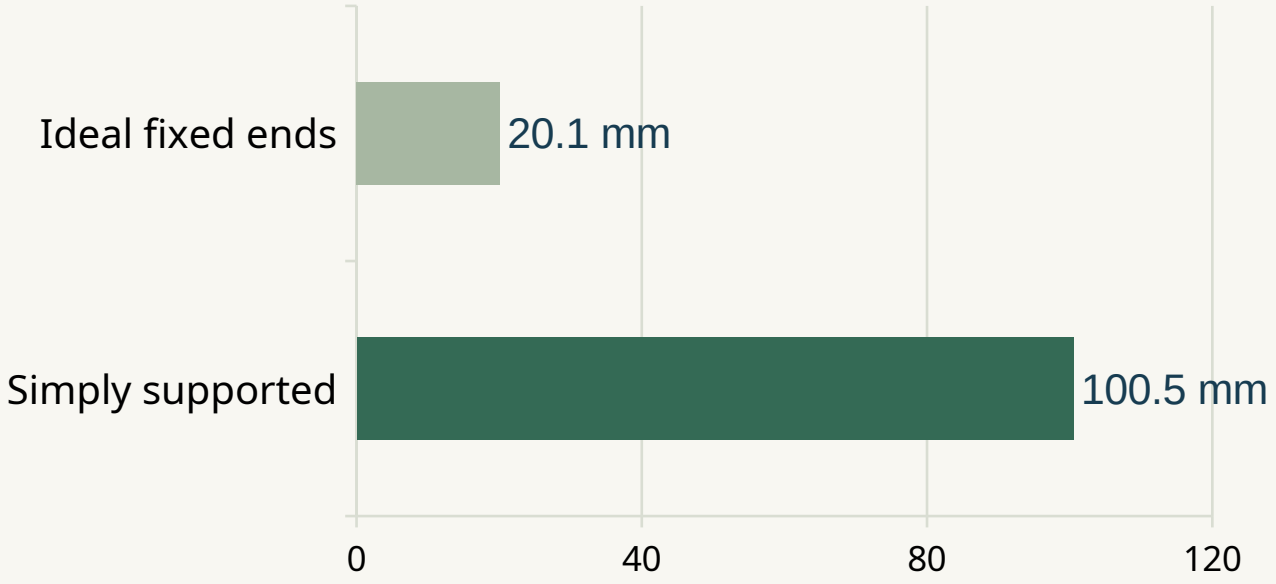
MATERIAL / SECTION	BENDING* kN·m	λ* W/(m·K)	CARBON* $kgCO_2e/kg$	MATERIAL COST* CNY / 288.66 m
Q355B tube Ø80×4	6.14	50	2.15	18,394
Q355B tube Ø80×3	4.78	50	2.15	13,983
Engineered bamboo Ø80	1.01	0.22	0.40	10,158
Treated timber Ø80	0.85	0.14	1.10	6,530
Raw bamboo Ø80	0.40	0.18	0.25	4,330

Steel: stiffness and protected outdoor use. Engineered bamboo: lower reference carbon/cost, but stiffness and treatment govern. Raw bamboo: low cost does not remove variability and care needs.

Source: material-comparison workbook (P3). *Preliminary inputs; product, processing and factor boundaries need verification.

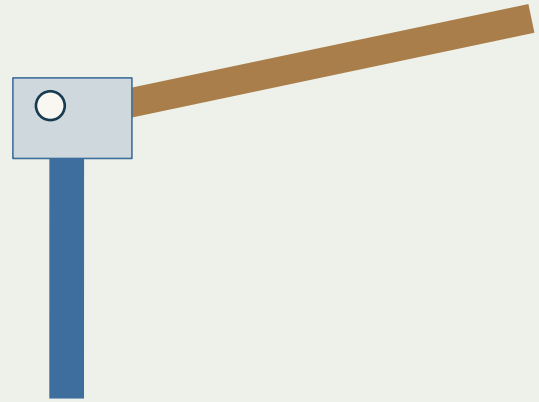
The crown is stiffness-sensitive: joints matter

Load assumption → boundary condition → deflection → required connection behaviour.



6.65 m SPAN · ADOPTED L/200 = 33.25 mm

PROPOSED SUPPORT-HEAD NODE



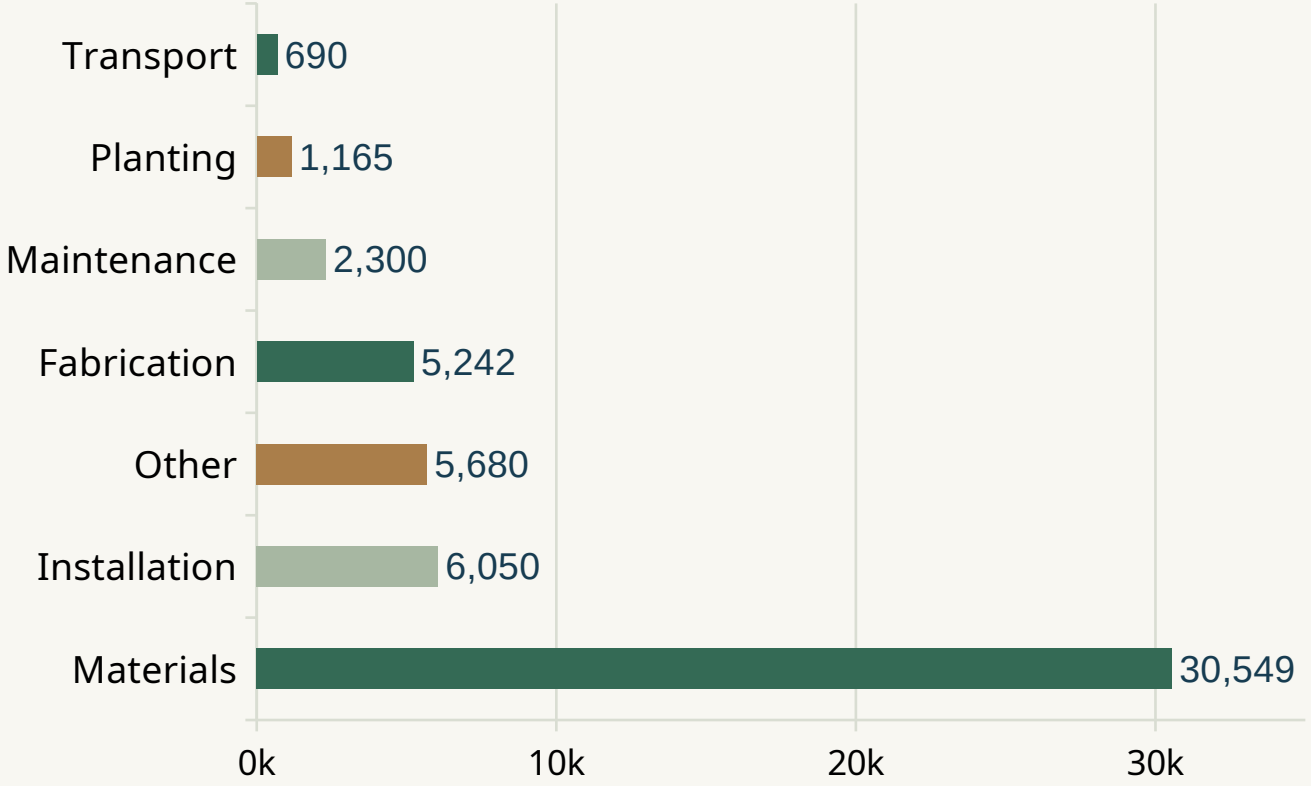
Node plate + sleeve + structural adhesive
with high-strength bolts

Concept only; rotational restraint must be verified.

Simplified beam calculation and proposed node method (P3). Wet foliage 18 kg/m² and bamboo properties are assumed inputs.

The unit budget includes fabrication, installation and care

Preliminary total: CNY 51,676 per unit—not simply the price of the members.



MATERIALS = 59.1% OF TOTAL

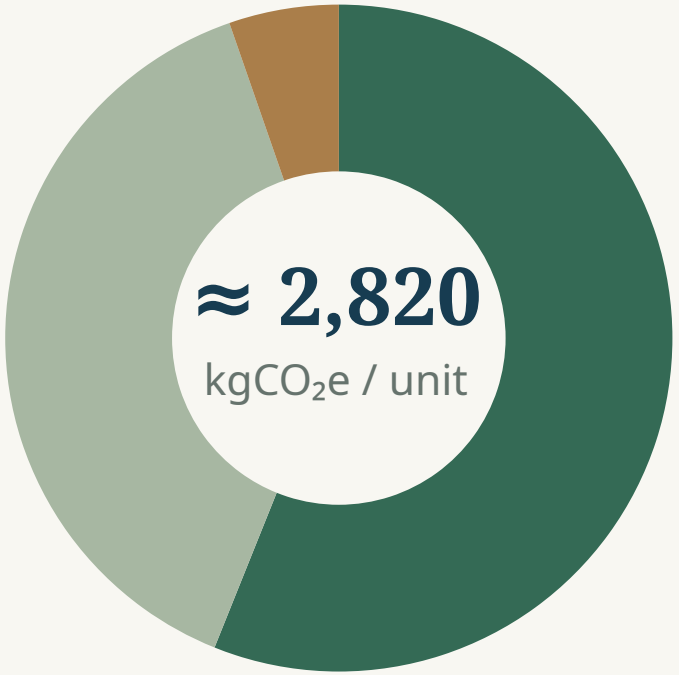
KEY COMPONENT	CNY
Bamboo crown	7,716
Steel supports	4,421
12 rigid-head nodes	12,000
Counterweight	3,402

These component rows are part of the full total. Nodes are a larger allowance than the primary steel members.

Source: Cost-Analysis.xlsx (P3). All rates remain reference estimates rather than confirmed quotations.

Account for the members, base and maintenance needs

Current carbon estimate: approximately 2,820 $kgCO_2e$ per unit.



CONTRIBUTOR	$kgCO_2e$
Primary members	1,581
Counterweight	1,089
Other components	150

Counterweight ≈ 39% of the current estimate

STEEL	BAMBOO	CONNECTIONS
Protect coating	Treat and inspect	Check fixings and grout

Workbook estimate (P3). Factor boundaries and node inclusion require checking; no biogenic-carbon credit is taken.

Select climbers for support, season and maintenance

The source proposal combines two Boston ivy plants with one Rangoon creeper per unit.

CANDIDATE / ROLE	CLIMBING METHOD	DESIGN CHECK
Boston ivy — primary	Tendrils with adhesive pads	Attachment and seasonal coverage
Trumpet creeper — alternative	Aerial roots	Vigorous growth and root control
Rangoon creeper — companion	Twining shoots	Local winter survival and leaf habit

2 + 1 PLANTS / UNIT

2 Boston ivy + 1 Rangoon creeper

ROOTS + WATER + TRAINING

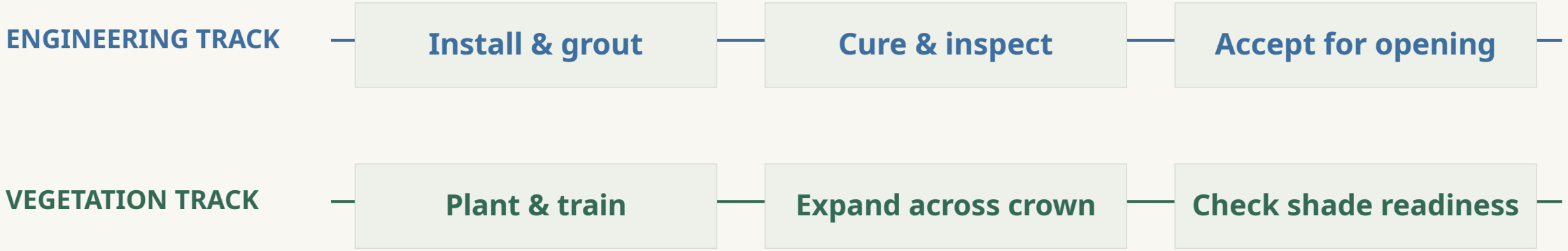
Source allowance: 1.2 m^3 substrate and a drip system.

Screening also considered wisteria (weight), honeysuckle and firecracker vine (winter leaf retention), and English ivy / star jasmine (leaf retention and public-space checks).

Project planting screen and schedule (P3): eight candidates. Proposed mix; local suitability, attachment and wet loads still need evidence.

Opening the structure is not the same as a mature canopy

Two readiness gates: structural acceptance and an agreed occupied-hour shade target.



TIME TO TARGET COVER: NOT YET ESTABLISHED

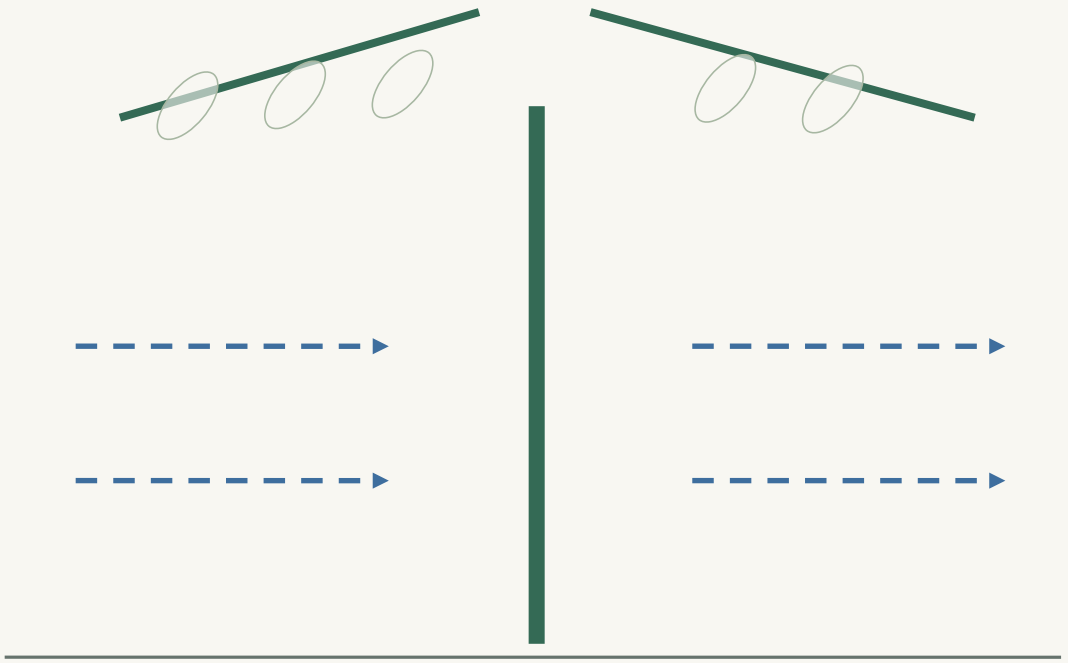
Month/year estimates need species, local climate, nursery starting size and maintenance evidence. The supplied 0–95% stages do not provide those durations.

Sequence only; intervals are not to scale. Neither 14 person-days nor a 3-year care budget predicts canopy maturity.

Source care/assembly proposals: P3, P4. Coverage stages: P5, not a calibrated biological timeline.

Different mechanisms need different evidence

The project calculates geometric/direct-beam shade; other responses remain to be validated.



CONCEPTUAL AIRFLOW

01 SHADING — ANALYSED

Reduce direct solar exposure of people and ground. [1,2]

02 EVAPOTRANSPIRATION

Depends on water, plant condition and weather. [2]

03 AIRFLOW — DESIGN INTENT

Open sides retain potential paths; dense foliage may add resistance.

Literature mechanisms: [1,2]. Airflow and application diagram: P4. No CFD, evapotranspiration balance or field cooling result is claimed.

Project shade onto one fixed reference zone

30°N · assumed X=east/Y=north · true solar time 09–15 · no surrounding obstructions.

01	SUN DIRECTION
02	MODEL + LEAF ENVELOPE
03	GROUND PROJECTION
04	TARGET INTERSECTION

IDEAL GEOMETRIC OVERLAP

Continuous-leaf shadow ∩ target ÷ target area

EQUIVALENT DIRECT-BEAM BLOCKAGE

$$A_{eq} = A_{bare} + (1 - \tau)(A_{leaf} - A_{overlap})$$

Divide Aeq by target area; τ = 0.15.

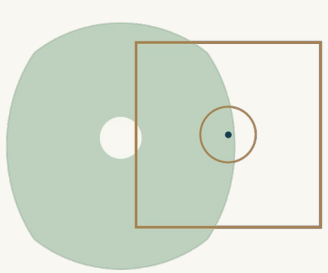
TARGET: 5 × 5 m MINUS r = 0.75 m → 23.23 m²

Not net usable floor area.

Solar geometry: NOAA [3]. Project calculations and scenario assumptions: P2, P5. Foliage/transmittance are not measured.

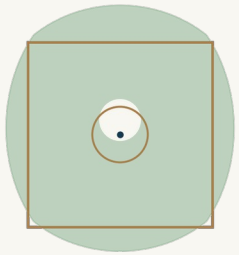
The shadow position and solar season both matter

Ideal continuous-leaf overlap: a high noon value does not mean continuous occupied-hour shade.



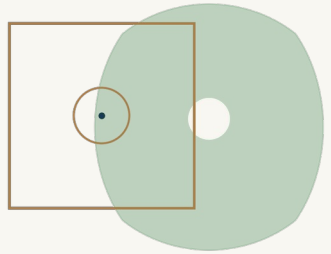
09:00

47.2%



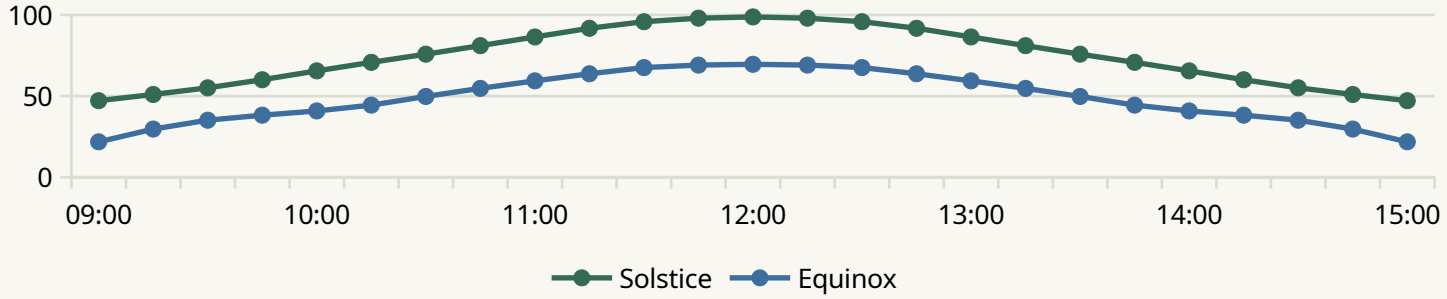
12:00

98.7%



15:00

47.2%



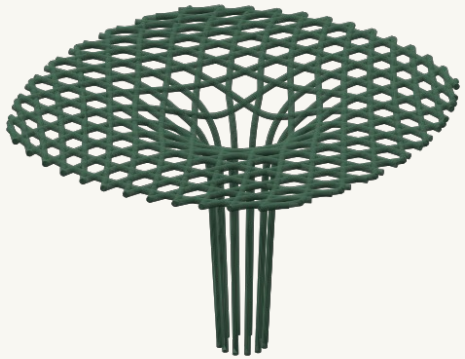
09–15 TIME MEAN

Solstice 75.3%
Equinox 49.8%

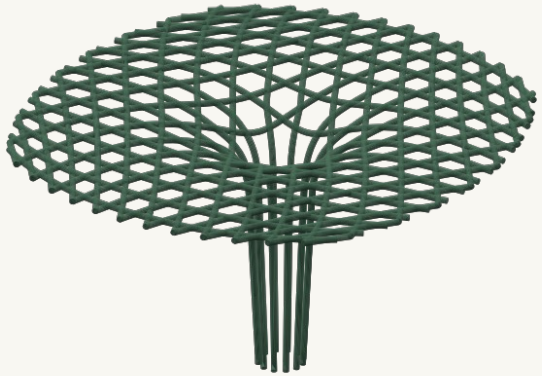
Project projection data (P2, P5). Solar-season comparison, not a calibrated seasonal plant-growth model.

Seven models test three geometry controls

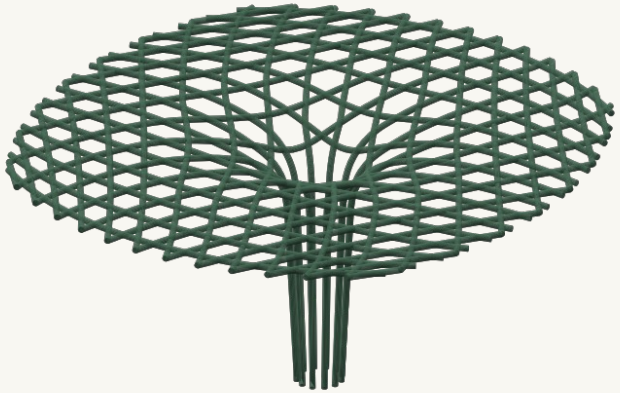
One variable at a time; same target, 50% plan cover, solstice solar hours and τ = 0.15.



SMALLER CROWN



BASELINE



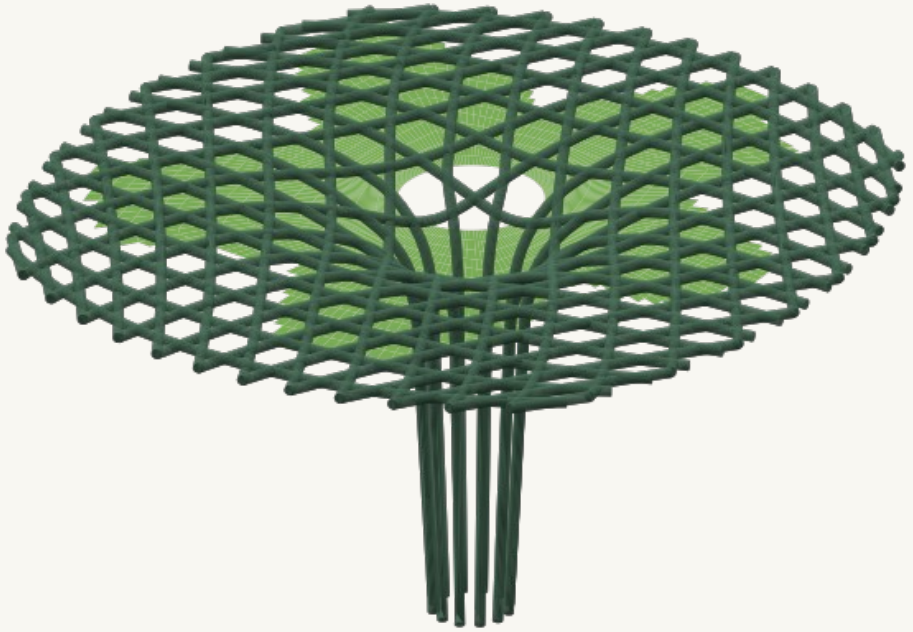
LARGER CROWN

CONTROL	TESTED SETTINGS	MEAN BLOCKAGE: LOW / BASE / HIGH
Upper expansion	0.92 / 1.08 / 1.24	52.3% / 60.2% / 66.1%
Upper-height control	−0.4 / 0 / +0.4 m	62.6% / 60.2% / 57.8%
Corner factor	0.55 / 0.85 / 1.00	61.8% / 60.2% / 59.1%

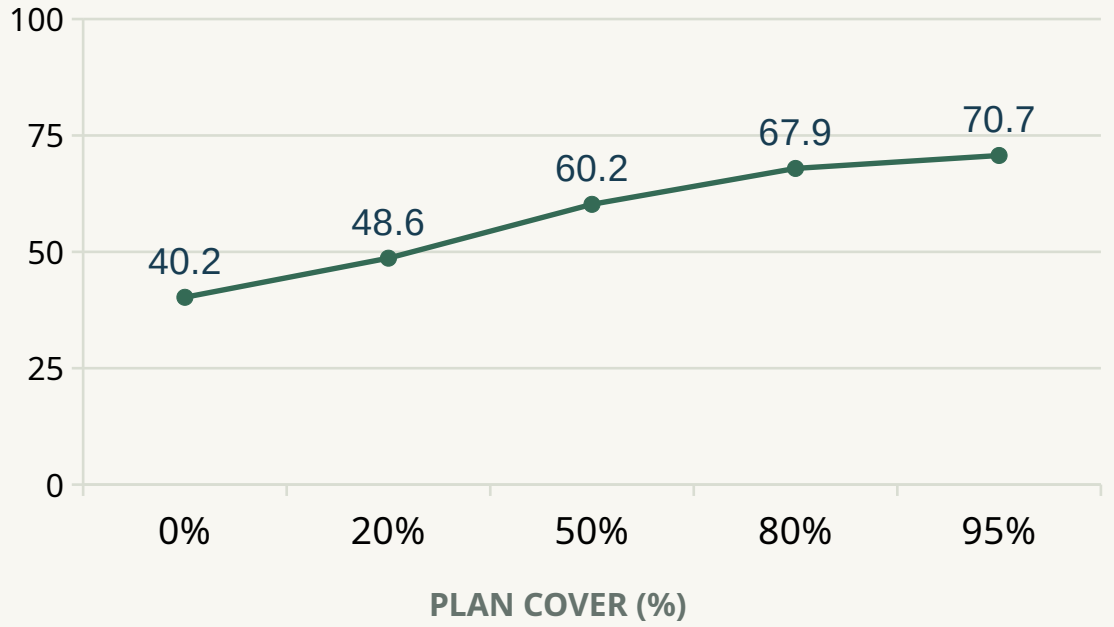
Project parameter study (P5). Controls are not uniform scaling; larger cover percentage does not imply equal plant/material input.

Coverage amount alone does not determine useful shade

Fixed final structure; five nested leaf-cluster scenarios, not observed growing months.



STATIC VIEW: 50% PLAN COVER • ANIMATION IN PPT

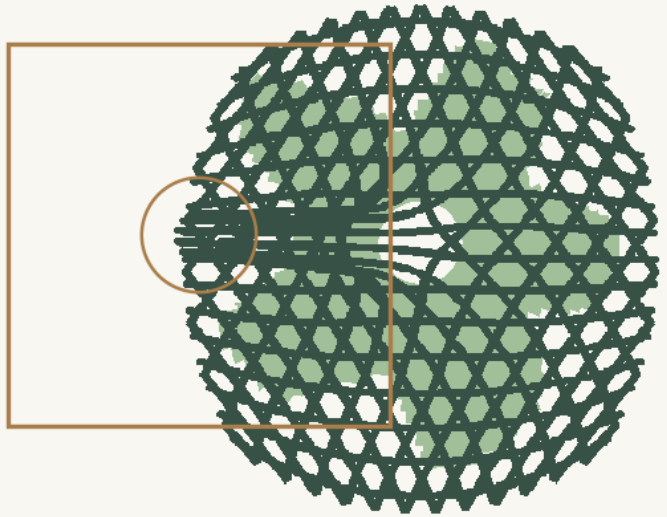


Same 50% cover, different layouts at noon: 77.5% / 71.4% / 62.6% blockage.

Project spatial-cover calculation (P5). Frame/leaf overlap deducted, τ = 0.15; equivalent direct-beam blockage, not cooling.

Put occupied-hour shade into the layout decision

A rest node must coordinate seating, through movement, roots and a substantial base.



STATIC VIEW: 15:00 • FIXED 50% PLAN COVER

01 SEATING LOCATION

Use the occupied-hour footprint, not just the crown outline or shadow centroid.

02 SPACE FOR ACCESS & ROOTS

Allow for circulation, counterweight, root zone, irrigation and maintenance.

03 VALIDATE THE OUTCOME

Confirm site and planting; compare radiation, temperature, humidity and wind.

Conceptual application P4; fixed-50%-cover projection P5. No real-site compliance or measured comfort result is claimed; method lead [4].

Coordinate prefabricated modules and site interfaces

Numbered components help assembly; long members, heavy base and grouted joints constrain it.

MODULE	WHAT IS PREPARED
Crown	47 formed/treatment-ready bamboo members
Supports	12 proposed galvanised steel members
Connections	Head nodes, sleeves, clamps and base flange
Foundation	Precast counterweight with anchorage
Planting	2 + 1 plants, substrate and drip provision

01 **Prefab, treat & number**

02 **Lift base and set out**

03 **Support, align & assemble**

04 **Grout, cure & recheck**

05 **Plant, train & restore**

14 person-days estimated ≠ 14 calendar days. The ≈9.07 t base controls lifting; grouted nodes limit lossless reversal.

Project member/assembly and cost schedule (P3, P4). Proposed detail and labour quantities, not a tested construction programme.

Connect geometry, construction and useful shade

Three design workstreams now have traceable outputs.

GEOMETRY

Rhino/GH model and parameters

Final form, drawings and material take-off

MATERIALS & BUILD-UP

Screening, sections, budget and modules

Proposed interfaces and implementation constraints

PERFORMANCE

Time-dependent shade and sensitivity

Geometry, cover and spatial-distribution effects

7 PARAMETER MODELS

5 COVER STATES

1,750 TIME-STEP RESULTS

Next: confirm products and joints, calibrate plant establishment, and test the real site.

Synthesis of project records P1–P5. Product, joint, site and growth-duration evidence are the next verification priorities.

References and project evidence

External method/mechanism sources are separate from project inputs and estimates.

[1] OUTDOOR SHADE & COMFORT

Watanabe et al. (2014). Evaluation of outdoor thermal comfort in sunlight, building shade, and pergola shade during summer in a humid subtropical region. Building and Environment, 82, 556–565.

[2] GREENING MECHANISMS

Hoelscher et al. (2016). Quantifying cooling effects of facade greening: Shading, transpiration and insulation. Energy and Buildings, 114, 283–290.

[3] SOLAR GEOMETRY

NOAA Global Monitoring Division. General Solar Position Calculations.
gml.noaa.gov/grad/solcalc/solareqns.PDF

[4] COMFORT METHOD

UTCI Calculator / COST Action 730.
www.utci.org/utci_calc.php
Method lead only; no project UTCI result is reported.

PROJECT RECORDS

P1 Model & take-off • P2 Ideal shade analysis • P3 Cost-Analysis.xlsx
P4 Proposed design & build-up • P5 Parameter and spatial-cover simulation

The growth-time page deliberately has no month/year curve without species- and site-specific evidence.